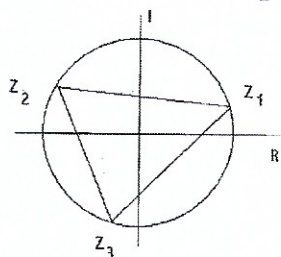


Complex Number 14

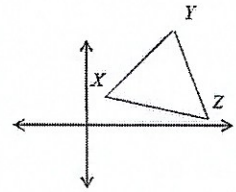
1. a. Use de Moivre's theorem to show that: $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$
 b. Hence show that the equation $16x^5 - 20x^3 + 5x - 1 = 0$ has roots $x = 1, \sin \frac{\pi}{10}, \sin \frac{9\pi}{10}, \sin \frac{13\pi}{10}, \sin \frac{17\pi}{10}$.
 c. By equating coefficients, or otherwise, find the values of b and c for which $16x^4 + 16x^3 - 4x^2 - 4x + 1 = (4x^2 + bx + c)^2$, and hence explain why the equation $16x^4 + 16x^3 - 4x^2 - 4x + 1 = 0$ has two double roots.
 d. Use part (b) to show that the equation $16x^4 + 16x^3 - 4x^2 - 4x + 1 = 0$ has roots $x = \sin \frac{\pi}{10}, \sin \frac{9\pi}{10}, \sin \frac{13\pi}{10}, \sin \frac{17\pi}{10}$
 e. Hence find exact values for $\sin \frac{\pi}{10}$ and $\sin \frac{3\pi}{10}$
2. a. Show that $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$
 b. Hence solve the equation $16 \sin^5 \theta = \sin 5\theta$ for $0 \leq \theta < 2\pi$
3. a. Use de Moivre's theorem to show that $\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$
 b. Hence show that the equation $x^4 - 10x^2 + 5 = 0$ has roots $x = \pm \tan \frac{\pi}{5}, \pm \tan \frac{2\pi}{5}$
 c. Deduce that $\tan \frac{\pi}{5} \tan \frac{2\pi}{5} = \sqrt{5}$ and that $\tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5} = 10$
4. Let $x = \cos \theta + i \sin \theta$
 a. Show that $2 \cos n\theta = z^n + \frac{1}{z^n}$ and that $2i \sin n\theta = z^n - \frac{1}{z^n}$
 b. Hence show that: $128 \cos^3 \theta \sin^4 \theta = \left(z^7 + \frac{1}{z^7}\right) - \left(z^5 + \frac{1}{z^5}\right) - 3\left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)$
 c. Deduce that $\cos^3 \theta \sin^4 \theta = \frac{1}{64}(\cos 7\theta - \cos 5\theta - 3 \cos 3\theta + 3 \cos \theta)$
5. a. Determine the range of $\arg(z)$, where $|z - 4i| = 2$.
 b. If $z = \cos \theta + i \sin \theta$ prove that $\frac{2}{1+z} = 1 - i \tan \frac{\theta}{2}$
 c. OABC is a square. $\overrightarrow{OA}$ represented by the vector $3 + i$. Find the vectors representing $\overrightarrow{OB}$ and $\overrightarrow{OC}$.
6. Using the fact that $|z|^2 = z\bar{z}$
 a. Show that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$
 b. i. Express $1 + \cos \theta + i \sin \theta$ in modulus argument form.
 ii. Find $(1 + \cos \theta + i \sin \theta)^2$ in modulus argument form.
 c. i. Express $\sin \theta + i \cos \theta$ in modulus argument form.
 ii. Hence find $\sqrt{\sin \theta + i \cos \theta}$ in modulus argument form.
7. Z_1, Z_2 and Z_3 are three points on the unit circle forming an equilateral triangle.
 a. If $Z_1 = \cos \theta + i \sin \theta$ express Z_2 and Z_3 in terms of θ and Z_1 .
 b. Prove that $Z_1 + Z_2 + Z_3 = 0$
 c. Prove that $Z_1^2 + Z_2^2 + Z_3^2 = Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1$



8. If ω is a complex cube root of unity
 a. Show that $(1 - \omega - \omega^2)(1 - \omega + \omega^2)(1 + \omega - \omega^2) = 8$.
 b. Form a quadratic equation with roots $a\omega + b\omega^2, a\omega^2 + b\omega$.

9. In the diagram below, the points X, Y and Z correspond to the complex numbers x , y and z respectively. Find the complex numbers represented by:

- The vector OX (where O is the origin).
- The vector XZ .
- The point A such that $XYAZ$ is a parallelogram.
- The point C , the centroid of $\triangle XYZ$. [Note: The centroid of a triangle is the point of intersection of the three medians. You may assume that the centroid lies two-thirds along a median from the vertex.]



10. Let α be the complex root of the polynomial equation $z^7 = 1$ with the smallest positive argument. Let $\theta = \alpha + \alpha^2 + \alpha^4$ and $\phi = \alpha^3 + \alpha^5 + \alpha^6$.

- Explain why $\alpha^7 = 1$ and $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$.
- Show that $\theta + \phi = -1$ and $\theta\phi = 2$.
- Show that $\theta = -\frac{1}{2} + i\frac{\sqrt{7}}{2}$ and $\phi = -\frac{1}{2} - i\frac{\sqrt{7}}{2}$.
- Show that $-\cos\frac{\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} = -\frac{1}{2}$.

11. Sketch (on separate diagrams) the region in the Argand diagram containing the points z for which:

- $\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{2}$ and $|z - 1 - 3i| \leq 2$
- $\arg\left(\frac{z-2i}{z+2}\right) = \frac{\pi}{4}$

12. a. Express $1 + i$ in modulus-argument form.

- b. Given that $(1 + i)^n = x + iy$, where x and y are real, and n is an integer, prove that $x^2 + y^2 = 2^n$.

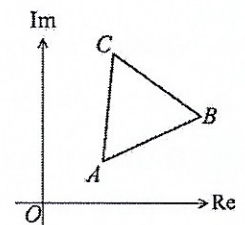
13. Find the smallest positive integer values of p and q for which $\frac{(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8})^p}{(\cos\frac{\pi}{12} - i\sin\frac{\pi}{12})^q} = i$

14. The cube roots of 1 are denoted by 1 , ω and ω^2 where $\text{Im } \omega^2 < 0$

- a. Show that $1 + \omega + \omega^2 = 0$

In the diagram ABC is an equilateral triangle. The points A , B and C represent the numbers z_1 , z_2 and z_3 respectively.

- Explain why $z_1 - z_3 = \omega(z_3 - z_2)$.
- Hence show that $z_1 + \omega z_2 + \omega^2 z_3 = 0$.



15. The complex number z satisfies the relation $|z + 4 - 4i| = 4$

- Sketch, on an Argand diagram, the locus of z .
- Show that the greatest value of $|z|$ is $4(\sqrt{2} + 1)$.
- Find the value of z for which $\arg(z + 4 - 4i) = \frac{1}{6}\pi$, leaving your answer in the form $a + ib$.

Answer

1. c. $b = 2, c = -1$ e. $\sin\frac{\pi}{10} = \frac{\sqrt{5}-1}{4}, \sin\frac{3\pi}{10} = \frac{\sqrt{5}+1}{4}$

2. b. $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

5. a. $\frac{\pi}{3} \leq \arg(z) \leq \frac{2\pi}{3}$ c. $\vec{OB} = 2 + 4i$, $\vec{OC} = -1 + 3i$

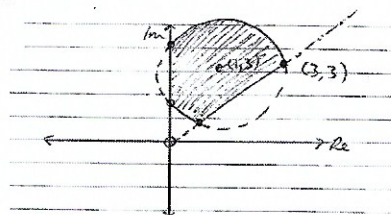
6. b. i. $2\cos\frac{\theta}{2}(\text{cis}\frac{\theta}{2})$ ii. $4\cos^2\frac{\theta}{2}(\cos\theta + i\sin\theta)$ c. i. $\cos(\frac{\pi}{2} - \theta) + i\sin(\frac{\pi}{2} - \theta)$ ii. $\pm \text{cis}(\frac{\pi}{4} - \frac{\theta}{2})$

7. a. $Z_2 = Z_1 \text{cis}\frac{2\pi}{3}, Z_3 = Z_1 \text{cis}(-\frac{2\pi}{3})$

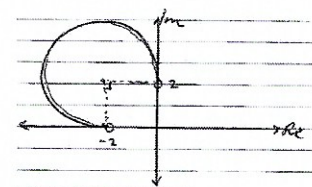
8. b. $z^2 + (a+b)z + (a^2 - ab + b^2) = 0$

9. a. x b. $z - x$ c. $y + z - x$ d. $\frac{x+y+z}{3}$

11. a.



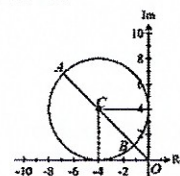
- b.



12. a. $\sqrt{2}\text{cis}\frac{\pi}{4}$

13. $q = 3, p = 2$

15. a.



- c. $z = 2\sqrt{3} - 4 + 6i$